Behind speech technology are two aspects that determine the voice-to-text flow. You have the speech synthesisers, which take in text as input and produce an audio stream as the output. These are generally referred to as text-to-speech (TTS) programs. You also have a speech recogniser, which is a program that turns the input and output streams of the TTS around: recognisers take in audio streams and output text.

For a synthesiser to work, it needs to first analyse the sting of characters that form the text, keeping a strong emphasis on grammar rules, natural language rules, and the language itself. English, for example, is a far easier language to deal with than, say, Japanese. The program thus needs to figure out multiple details on the text: which words are proper nouns, numbers, exclamation points; where sentences begin and end; whether a phrase is a question or a statement; and whether a statement is in the past, present, or future tense. These elements are all vital in determining the final output—the appropriate pronunciations and intonations for the words and phrases that form the sentence.

The system must then analyse and generate appropriate sounds for the processed text. It needs to determine where to add emphasis, where to break, or pause, mid-sentence, and so on. Finally, it has to actually speak out the sentence—this was done algorithmically in the past, which generated the typical "computer-sounding" voice that is common to such solutions. Windows Vista utilises a database of sound segments built from hours and hours of recorded speech. The program thus selects the appropriate sounds from this database and then splices these various sound segments together to speak the sentence.

Speech *recognition* is vastly more complicated than synthesis. You can imagine the scale of complication when you realise that the input is essentially a sound wave that needs to be analysed and processed—to understand everything from intent to placement within a statement. A speech recognition program needs to process the audio stream and isolate the segments that it recognises as actual speech